

Numerical Analysis

Lab 11

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This lab deals with numerical differentiation and solution of ODEs.

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1 One step methods for ODEs

One step methods for ODEs I

We consider Cauchy problems of the type:

$$\begin{aligned} &\text{find } y \in \mathcal{C}^1(I) \text{ such that} \\ &\quad \begin{cases} y'(t) = f(t, y(t)) & t \in I \\ y(t_0) = y_0. \end{cases} \end{aligned}$$

One step methods for ODEs II

Forward Euler method

$$y_{n+1} = y_n + hf_n$$

Backward Euler method

$$y_{n+1} = y_n + hf_{n+1}$$

Crank-Nicolson method

$$y_{n+1} = y_n + \frac{h}{2} [f_n + f_{n+1}]$$

θ -method

$$y_{n+1} = y_n + h[\theta f_n + (1 - \theta)f_{n+1}]$$

A method is **explicit** if y_{n+1} is computed in terms of previous values y_k , $k \leq n$. If y_{n+1} depends implicitly on itself through f the method is **implicit**.

One step methods for ODEs III

Unitary Local Truncation Error

The unitary local truncation error is defined as

$$\tau_{n+1}^h = \frac{y(t_{n+1}) - u_{n+1}^*}{h}.$$

Order

A method is said of order p if

$$\tau_{n+1}^h = \mathcal{O}(h^p).$$

Consistency

A method is consistent if

$$\lim_{h \rightarrow 0} \tau_{n+1}^h = 0.$$

Convergence

A method is convergent if

$$\lim_{\substack{h \rightarrow 0, \\ nh=t}} y_n = y(t) \quad \forall n.$$

Euler methods are of first order. Crank-Nicolson method is of second order.

Euler methods

$$\tau_{n+1}^{(h)} = \frac{h}{2} y''(\xi) \quad \xi \in (t_n, t_{n+1})$$

Crank-Nicolson method

$$\tau_{n+1}^{(h)} = \frac{h^2}{6} y'''(\xi) \quad \xi \in (t_n, t_{n+1})$$

Exercise 11.1

Consider the following Cauchy problem

$$\begin{cases} y' = t - y + 1 & 0 \leq t \leq 1 \\ y(0) = 1 \end{cases}$$

whose solution is $y(t) = t + e^{-t}$.

- a Implement the Forward Euler method in a suitable function.
- b Determine the step size h needed by the Forward Euler method to approximate the solution in $T = 1$ with an absolute error smaller than 0.01. Numerically verify the result.

Exercise 11.2

Consider the following Cauchy problem

$$\begin{cases} y' = -ty^2 & 0 \leq t \leq 2 \\ y(0) = 1 \end{cases}$$

whose solution is $y(t) = \frac{2}{t^2 + 2}$.

- a Implement the Backward Euler method in a suitable function, solving the nonlinear equation for y_{n+1} .
- b Determine the step size h such that the unitary local truncation error after the first step is smaller than 10^{-3} . Solve the problem using such step.

Exercise 11.3 (Exam July 20, 2023)

Consider the following Cauchy problem

$$\begin{cases} y' = -\frac{1}{t}(2y + t^2 + y^2) & t \in [1, 5] \\ y(0) = 1 \end{cases}$$

- a Let us approximate the given Cauchy problem using the Matlab function `forward_euler.m` with $h = 0.2$. Report the Matlab commands used and the values of the computed numerical solution.
- b Given that the exact solution is $y_{\text{es}}(t) = t^2(1 + \log(t))$, represent on the same graph the exact solution, the numerical solution computed in point a, and the numerical solution computed using the forward Euler method with $h = 0.1$. Upload the resulting graph in .png format.
- c Report the Matlab commands used and comment on the results obtained in point b.
- d Compute the numerical solution u_n for $n = 0, \dots, N$ of the given Cauchy problem with the forward Euler method and $h = [0.1, 0.05, 0.025, 0.0125]$. Compute the error $e_h = \max_{0 \leq n \leq N} |y(t_n) - y_{\text{ex}}|$, with $t_n = 1 + nh$ $n = 0 \dots N$ and $N = 4/h$. Report the error in a loglog plot.
- e Report the Matlab commands used in the computation of e_h and the values of e_h . Comment on the results obtained in point d in light of the theory.

Exercise 11.4 (Exam July 11, 2024)

Consider the following Cauchy problem

$$\begin{cases} y' = e^{-t} \sin(y(t)) & 0 \leq t \leq 1 \\ y(0) = \pi/2 \end{cases}$$

Exam July 11, 2024

- a Let us approximate the given Cauchy problem using the Matlab function *backward_euler.m* with $h = 0.1$ and $h = 1$. Report the Matlab commands used.
- b Upload the plot of the solutions obtained in point a in a unique graph in .png format. Use hold on command and legend to distinguish the solutions. (Hint: use help to plot the curve with different colors and markers).
- c Comment the obtained results in point b regarding absolute stability, accuracy and computational complexity.
- d Solve the same problem of point a with the forward Euler method and $h = 2$. Report the Matlab commands used and the values of the solution at $t = 2$ and comment the results.

Exercise 11.5 (Exam June 17, 2024)

Consider the following Cauchy problem

$$\begin{cases} y' = -\frac{t+1}{t+2}y(t) & 0 \leq t \leq 1 \\ y(0) = 2 \end{cases}$$

The exact solution is $y_{\text{ex}}(t) = (t+2)e^{-t}$.

Exam Sep 6, 2024

- a Use the Matlab function `crank_nicolson.m` to approximate problem 1 using the Crank Nicolson method with step size $h = 0.5$. Note that the function `crank_nicolson.m` internally uses the Newton method. Report the approximations of $y(1)$ and $y(5)$, along with the Matlab commands used. Finally, provide the number of iterations performed by the Newton method to compute $y(1)$ and $y(5)$.
- b Use Matlab to graphically represent the exact solution $y_{\text{ex}}(t)$ over the interval $[0, 5]$, and overlay it on the graph of the numerical solution. Represent the exact solution with a solid line and the numerical solution with a dashed line. Upload the resulting graph.
- c Use the Crank Nicolson method to solve the problem with step size $h_1 = 1$, $h_2 = 0.5$, $h_1 = 0.25$, $h_1 = 0.125$ and compute the value of the errors:

$$e_h = \max_{0 \leq n \leq N} |y(t_n) - y_{\text{ex}}(t_n)|$$

Report the values of e_h and the Matlab commands used.

- d Use the obtained results to estimate the converge order of the method with respect to h . Comment on the results obtained in point d in light of the theory.

Exercise 11.6

Consider the Cauchy problem

$$\begin{cases} y' = -2xy & 0 \leq x \leq 1 \\ y(0) = 1 \end{cases}$$

with exact solution $y(x) = e^{-x^2}$.

- a Starting from the function for the Backward Euler method, implement the θ -method.
- b Consider the cases $\theta = 0$ (BE), $1/4$, $1/2$ (crank nicolson), $3/4$ and 1 (FE). Apply the method with $h = 2^{-n}$ for $n = 4, \dots, 8$ and approximate the convergence order.

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2 Homework

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